RNA-Protein Interaction Classification via Sequence Embeddings

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In a nutshell

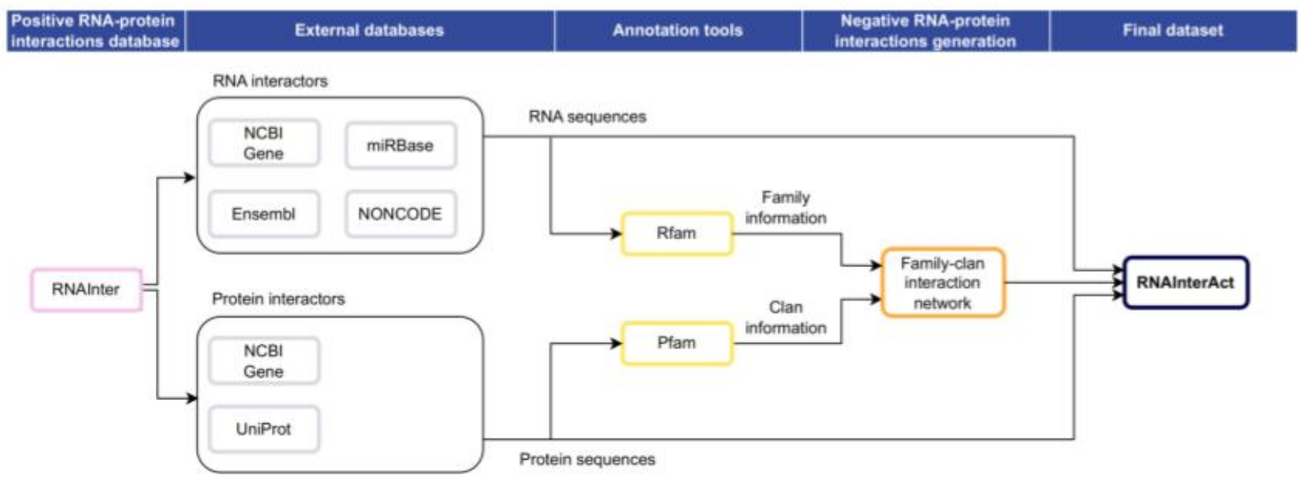
- RNA and proteins form complexes through **binding interactions**, crucial for **advances in biology or medicine**, such as drug development.
- Predicting RNA-protein interactions (RPIs) computationally is challenging due to **data scarcity** and its **bias**.
- We present **RNAInterAct,** the largest publicly available dataset with **biologically plausible negative interactions**.
- The dataset includes **sequence information** and precomputed **embeddings that capture meaningful information** about interactors' structure and function.
- A meticulous train/test split based on the **RNA family** enables evaluation on truly unseen interactions.
- We introduce **RPIembeddor**, a transformer-based model for binary interaction classification and demonstrate its strong performance.

RNAInterAct

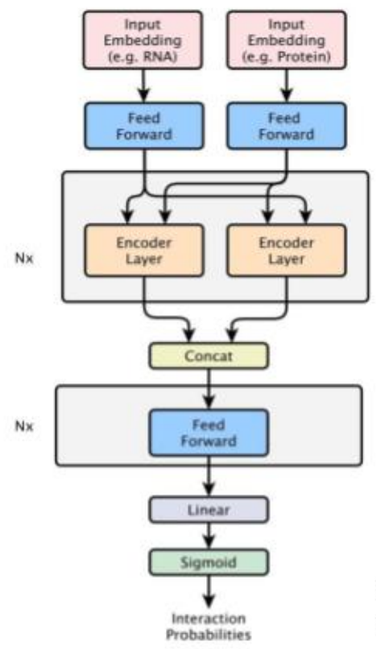
Compilation of the RNAInterAct dataset in five steps:

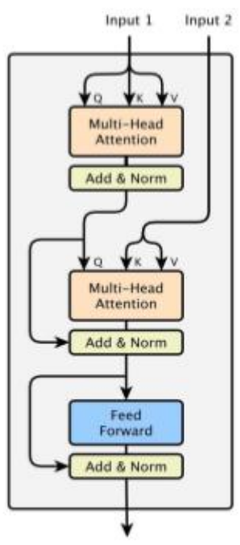
1. **Gathering positive RPIs** from the RNAInter[1] database:
2. Cross-linking large scale databases to **obtain sequence information**.
3. **Filtering**.
4. **Annotating** proteins with clan and RNA with RNA family information.
5. **Generating biologically relevant negative interactions**.

The final dataset contains **122,217 RPIs** with a **1:2 positive to negative ratio**.



RPIembeddor





- **Protein embeddings:** ESM-2[3].
- **RNA embeddings:** RNA-FM[4].

RNA family-based train/test split

- Dataset split into **TRinter** (training) and **TSfam** (test) sets **based on RNA families**.
- **Non-homologous data** guarantees evaluations on truly unseen interactions.

Feature	TRinter	TSfam	RPI2825[2]
Unique RNA Families	976	172	N/A
Unique Protein Clans	152	152	N/A
Positive Interactions	35,852	4,887	871
Negative Interactions	73,362	8,116	0
Total Interactions	109,214	13,003	871

Ablation studies

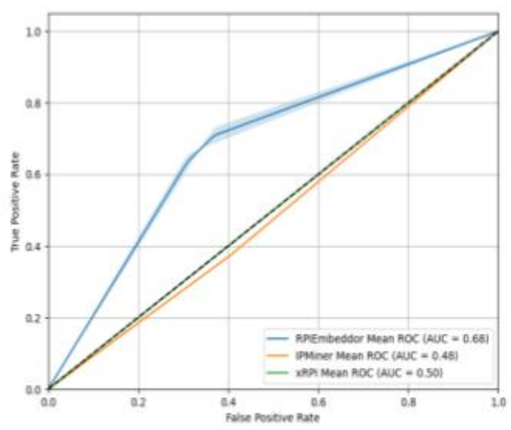
Model	Prec.	Rec.	F1	Acc.
RPIembeddor	**0.563±0.019**	**0.659±0.071**	**0.605±0.019**	**0.678±0.009**
One-Hot	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0
Random-Protein	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0
RNA-Random	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0

Both embeddings contribute to RPIembeddor's performance.

Evaluation on TSfam and RPI2825

Model\Test set	TSfam				RPI2825			
	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1	Acc.
RPIembeddor	**0.550 ±0.010**	0.627 ±0.017	**0.586 ±0.013**	**0.667 ±0.009**	**1.0 ±0.0**	0.667 ±0.085	0.8 ±0.049	0.667 ±0.085
IPMiner	0.357	0.375	0.366	0.512	**1.0**	0.107	0.193	0.107
XRPI	0.375	**0.909**	0.531	0.398	1.0*	0.982*	0.991*	0.982*

*XRPI has been trained on the RPI2825 dataset.



- **RPIembeddor generalizes across data distributions.**
- **Competitors behave like random classifiers.**

[1] Juanjuan Kang et al. "RNAInter v4.0: RNA interactome repository with redefined confidence scoring system and improved accessibility." Nucleic Acids Research, 2022. doi: 10.1093/nar/gkab997.

[2] Dharm Skandh Jain et al. "A data-driven model for predicting RNA-protein interactions based on gradient boosting machine." *Scientific Reports*, 8(1):9552, 2018.

[3] Z. Lin et al. "Evolutionary-scale prediction of atomic-level protein structure with a language model." *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574. Epub 2023 Mar 16.

[4] Jiayang Chen et al. "Interpretable RNA foundation model from unannotated data for highly accurate RNA structure and function predictions." *arXiv preprint* arXiv:2204.00300, 2022.

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Paper
https://openreview.net/forum?id=04MgOkuPTo

Dataset
https://ml.informatik.uni-freiburg.de/research-artifacts/RNAInterAct/RNAInterAct.tar.gz